



Project Title: SafePlate: Inferring Dietary Safety through Ingredient-Allergen Knowledge Graphs

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Mode: ☒ 6 ECTS / ☐ 3 ECTS

Motivation. Food allergies are a major public health concern, yet many online recipe platforms lack systematic allergen labelling. This is particularly dangerous for "hidden" allergens where an ingredient name does not explicitly state its category (e.g., 'Whey' for someone with a 'Dairy' allergy).

Problem. Most recipe data exists in "flat" formats—simple lists of ingredients with no semantic connection to medical or dietary knowledge. In this project, the problem we want to solve is to build a Knowledge Graph that can automatically detect allergens in unlabelled recipes by understanding the underlying properties of their ingredients.

Solution. The envisioned solution consists of a logic-based KG component that maps ingredients to allergen categories using a multi-hop path (Meal → Ingredient → Allergen). This will be combined with a KG embedding-based component to find safe alternatives by identifying ingredients that are used similarly but lack the target allergen's "creative DNA" in the graph.

Steps. The Plan is to use

- A curated subset of the Food.com recipe dataset (providing Meal → Ingredient links) and combining it with the Kaggle 16-tag allergen taxonomy to form the data of our Knowledge Graph.
- On top of that, the plan is to formulate logical knowledge representing safety rules (e.g., contains (Meal, Ingredient) & is(Ingredient, Allergen) -> unsafe(Meal, Allergen)) and use a logical reasoner to form the knowledge component of the KG.
- As a KG embedding component, the plan is to use a standard library (like TransE or RotatE) to enable ingredient substitution services.

Scope. The project will focus on a subset of approximately 2,000 recipes to ensure high data quality. We will handle Entity Resolution by matching ingredient strings from the recipes to



the allergen taxonomy using normalization and keyword matching. The plan is to use KG embedding methods out-of-the-box without major architectural adjustments.

Learning outcomes I plan to focus on	LO2, LO6
Learning outcomes I plan to show basic proficiency in	LO1, LO4, LO5, LO7, LO8, LO9, LO10, LO11, LO12
Learning outcomes I plan not to include	LO3